

Portfolio run, October 2026

Oct 6, 2026 · @Casey

Verdict

The book you control holds about 3.4 independent bets against Dalio's 15. Its two biggest imbalances are fixable without leverage: early-stage venture carries two-thirds of the risk, and nothing in the book pays if growth disappoints.

1. **What is working.** The liquid book holds about 3.8 independent bets, and some of its streams are genuine diversifiers: the Crash Convexity sleeve made money on stocks' worst days, and KMLM trend and the ETH carry have near-zero correlation to everything else.
2. **Change first, no leverage needed.**
 - Cap BTC at about 20% of liquid risk.
 - Put about \$250k of spare cash into long Treasuries (+0.4 independent bets).
 - Add about \$100k to KMLM trend, which is uncorrelated with everything you hold.
3. **Make the return engines "good" in Dalio's sense.** Private credit and venture are your alpha streams, which fits his idea, but today they are concentrated:
 - Film lending is 2.2 effective loans, the largest is 37% of the book, and realised income is 19% of plan.
 - Venture without Organics Ocean models as 1.35 effective bets.
 - The fix is more, smaller loans across borrowers, and equal-sized venture cheques (DexMat at \$75k fits).
4. **Treat Organics Ocean as a dated event, not a holding to rebalance.** Protect the exit, consider selling a slice early, and check the tax timing of a July 2027 close.
5. **After the sale, don't rebuild one bet.** The \$18M plan as written puts all its risk in US stocks and Composer. Your credit-and-venture route is the right shape. Move risk budget from Composer into a balanced core, prove the film return before scaling it, and accept about 7–11% volatility instead of borrowing to reach 15%.

Scorecard

Dalio's target is 15 independent bets; liquid asset classes alone offer about 2–3. Box balance runs from 0 (all risk in one growth/inflation environment) to 1 (equal risk in all four).

Book	Value	Volatility	Independent bets (DR ²)	Largest risk	Falling-growth box	Box balance
The book you control (without Organics Ocean)	\$7.36M	16.3%	3.4	Venture 69%	0%	0.45
Liquid book	\$2.37M	10.8%	3.8	BTC 41%	–1%	0.61
Liquid book, unlevered balanced	\$2.37M	4.9%	5.3	Five sleeves at 15–16%; BTC 12%	14%	0.83
\$18M plan as written	\$18M	10.4%	2.2	Stocks 51%	0%	0.33
\$18M balanced, unlevered	\$18M	7.2%	4.0	Stocks 24%	14%	0.73
\$18M your route (credit + venture)	\$18M	10.6%	4.3	Composer 41%	3%	0.41

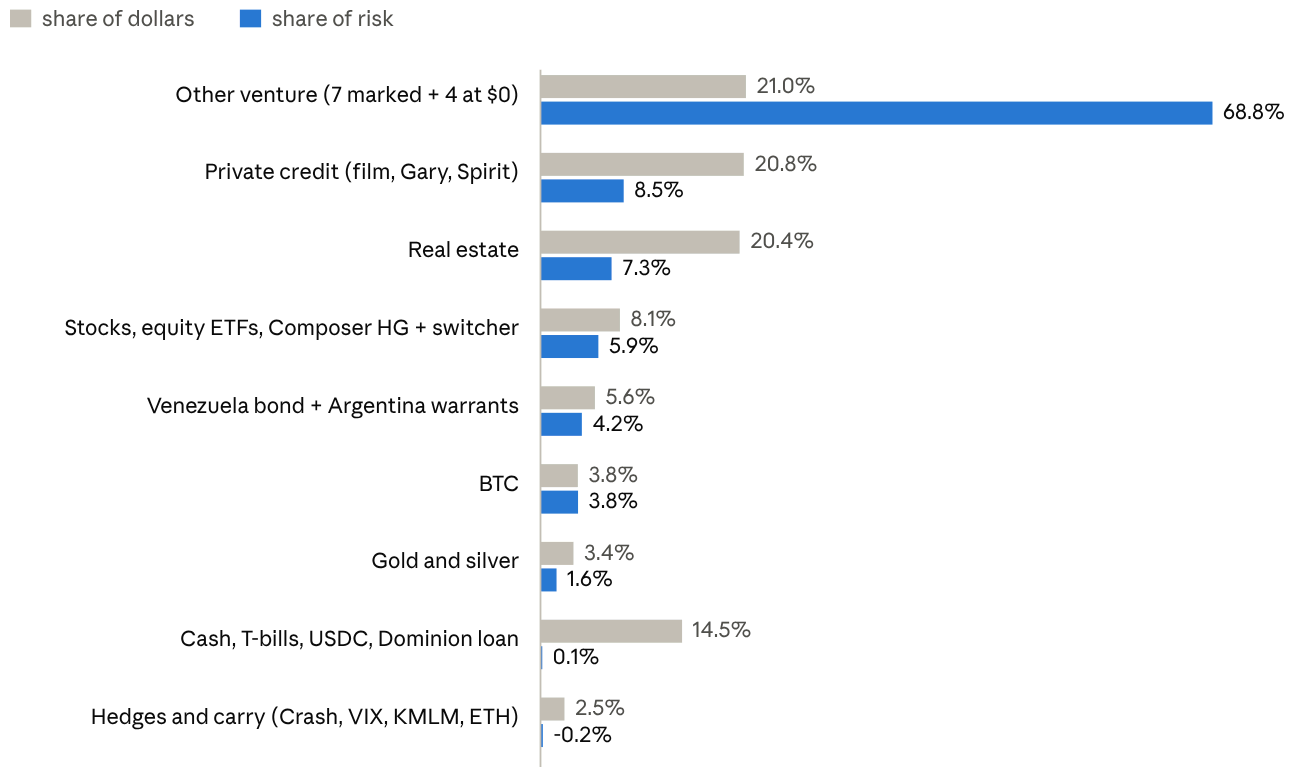
Whole book, for completeness: \$19.4–19.6M after the loan correction, with 1.5 independent bets and Organics Ocean 95% of risk on a volatility model (measured before the correction, which only moves cash; see its section). Liquid figures assume the Dominion loan is repaid into BOXX, its scheduled state; while it is out, 10.9% volatility and 3.9 bets.

Where risk sits

The book you control is \$7.36M, runs at about 16% volatility and holds 3.4 independent bets. Two-thirds of its risk sits in the early-stage venture names, and none in the falling-growth box.

Without Organics Ocean, venture is 21% of your dollars and 69% of your risk

The book you control, \$7.36M: share of dollars vs share of volatility (Euler contributions)

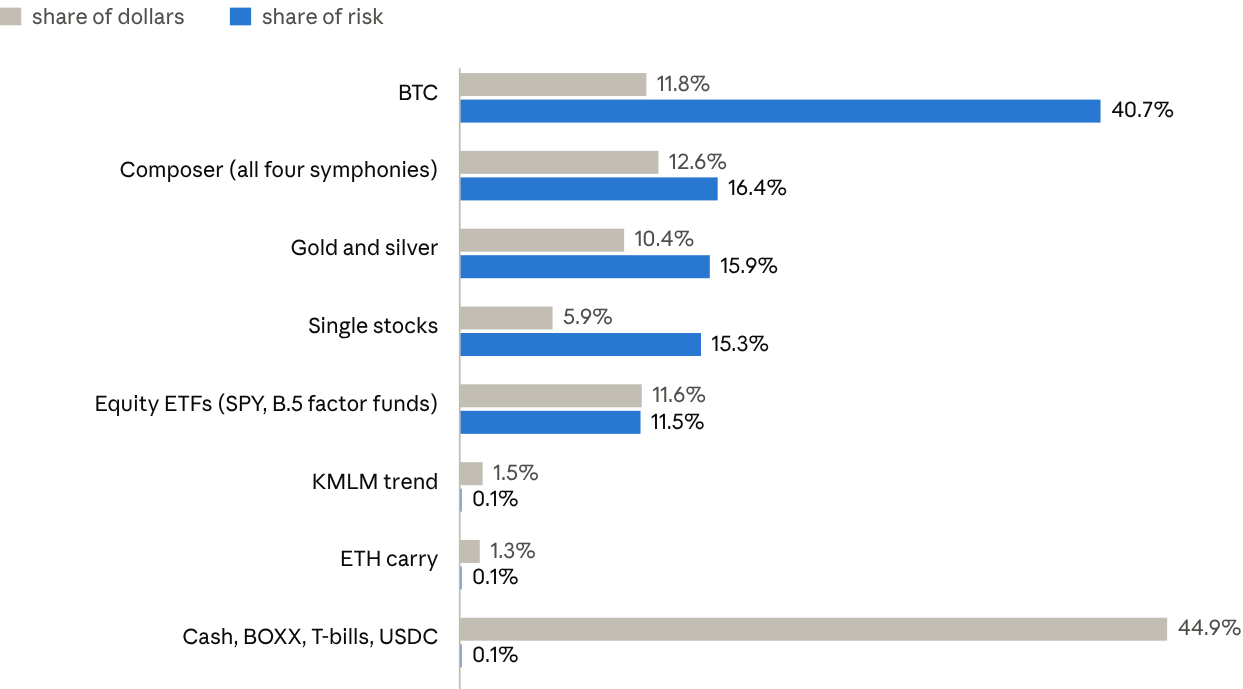


holygrail-engine results/casey-2026-10/controllable · ex_oo_risk_vs_dollar.csv · Dominion loan outstanding · independently recomputed

Venture's share rests on its assumed volatility, the softest input here: 42% of risk at 30% volatility, 58% at 45%, 69% at 60% (the base case) and 78% at 80%. Independent bets run from 4.5 to 2.9 across that range. Counting Series X at the \$184.8k actually funded, not the \$400k commitment, venture is 63% of risk. Every version puts venture first, and the falling-growth box empty.

In the liquid book, BTC is 12% of the dollars and 41% of the risk

Liquid book, \$2.37M (Dominion loan back in BOXX): share of dollars vs share of volatility



holygtrail-engine results/casey-2026-10/no-leverage · n2_risk_vs_dollar_by_sleeve.csv · independently recomputed

Inside the liquid book, BTC alone carries 41% of the risk, while Composer, gold and silver, single stocks and equity ETFs carry 11–16% each. Cash is 45% of the dollars and none of the risk.

Organics Ocean: a sale-pending position

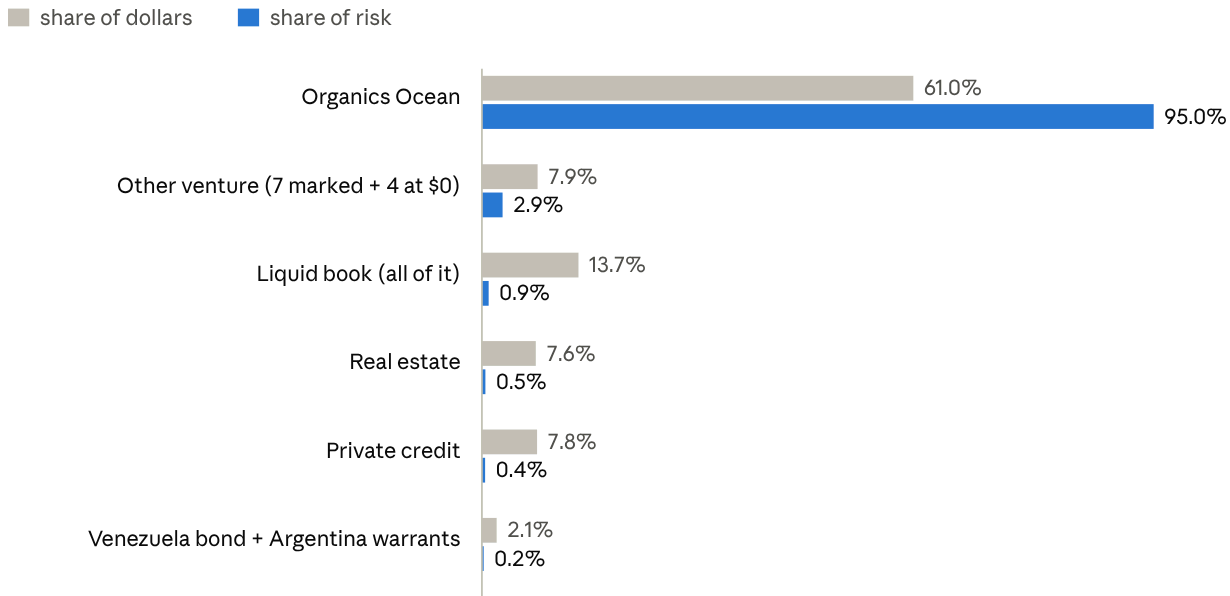
Organics Ocean was a \$100k venture cheque, is now marked at about \$12M, and has already returned \$254,900 in cash. It sits outside the scorecard because you cannot easily resize it before the sale targeted for July 2027. What matters is the range of outcomes on that date.

Outcome by July 2027	Value	Change vs the \$12M mark	Your odds
Sells at \$15–20M (your target is \$20M)	\$15–20M	+\$3M to +\$8M	to fill in
Sells within the bankers' range	\$10–15M	–\$2M to +\$3M	to fill in
Slips or the process fails	to fill in	to fill in	to fill in

- **Why it still dominates.** Even the bankers' own \$10–15M range ($\pm \$2.5M$) is about ten times the liquid book's one-standard-deviation year (about \$260k). Treated as a traded stock, it is 95% of the whole book's risk:

Organics Ocean is 61% of your net worth and 95% of its risk

Share of dollars vs share of volatility (Euler contributions), whole book at mark before the loan correction, \$19.7M



holylgrail-engine results/casey-2026-10/current · a1_whole_risk_vs_dollar.csv · volatility model; Organics Ocean at 40–100% vol gives 88–98%

- **Protect the exit, not the mark.** A sale price depends on risk appetite. A market drop before July could cut the price or delay the deal just as the liquid book falls. Which assets pay depends on the drop: long Treasuries rose in 2008 but fell 23% in 2022, when inflation drove the sell-off and commodity producers rose. Covering both kinds is what the four-box balance is for; the Crash Convexity sleeve is your one hedge that paid on stocks' worst days.
- **Take chips off if you can.** A partial secondary sale before July would turn part of a pending outcome into a settled one.
- **Check the date with your tax adviser.** Your sheet dates the investment Q4 2022, which would reach QSBS's five-year mark in Q4 2027, and Act 60 sourcing may also matter. Taxes are not modelled here.

What is working

Your liquid book holds about 3.8 independent bets. Some of its streams have genuinely different drivers:

Stream	Size	Correlation to the rest of the liquid book	Correlation to SPY	Average move on SPY's worst 5% of days
Composer Crash Convexity sleeve	\$77.9k	-0.18	-0.25	+2.0%
KMLM trend ETF	\$34.4k	0.04	-0.05	0.0%
Hyperliquid ETH carry (modelled)	\$30.1k	0.03	0.02	—
Composer KMLM switcher	\$97.4k	0.15	0.28	+1.0%
Composer VIX Harvester	\$37.4k	0.18	0.11	-0.1%
Physical gold and silver	\$170.7k	0.38	0.27	-0.9%

SPY's worst 5% of days averaged -2.2%.

- **The Crash Convexity sleeve is your best hedge.** It made money on the days stocks fell hardest and lowers total risk (a negative risk share).
- **Trend and carry are near-zero-correlation streams**, exactly the kind Dalio means. They are small: together under 3% of the liquid book.
- **Gold and silver cover the inflation box**, which most portfolios leave empty.
- **Private credit diversifies on paper.** Film and bridge loans are modelled at about 0.2 correlation to stocks, but that is an assumption, not a measurement; their quality is examined below.

Basis: daily returns, trailing 3 years to 2026-09-30. Composer series are backtests before each symphony's live start, so their correlations are partly in-sample.

What needs work

Five problems. None depends on a return forecast; the first four are measured, and the fifth rests on modelled default and venture correlations.

- **BTC carries 41% of the liquid book's risk on 12% of its dollars.** That holds in every version of the corrected book. One position should not set the liquid book's year.

- **Nothing pays if growth disappoints.** The falling-growth box holds –1% of risk: the book owns no long Treasuries or other duration. A recession with falling inflation would hit stocks, BTC, Composer and the exit price at once.
- **Composer is mostly one equity bet, and it is levered inside.** The HG symphony (not Dalio's Holy Grail) has 0.59 correlation to SPY and 0.56 to QQQ, and the KMLM switcher holds 3× equity ETFs when risk-on. There is no margin anywhere: exposure is 0.55× the liquid book at face value, and at most 0.8× if every Composer dollar ran at 3× (an upper bound).
- **About 45% of the liquid book sits in cash and T-bills.** That is why it runs at about 11% volatility. Cash is not a bet in Dalio's sense; it is idle risk budget. Net of the \$410k already committed (Series X, DexMat, Jarod Production), roughly a third is cash.
- **The return engines are concentrated.**
 - Film lending: \$1.48M in 7 loans, the largest (12 Dates of Christmas) 37% of the book. At the modelled 0.3 default correlation, that is 2.2 effective loans. Realised income over the last 12 months is 19% of plan, and Father of Us is past due (\$10k of the \$50k profit collected).
 - Venture: Organics Ocean is 97% of the paper gain. Without it, the remaining venture names model as 1.35 effective bets (they are modelled as one basket); even 17–18 new equal-sized deals would reach only about 2.9 at a 0.3 correlation.

Basis: daily returns to 2026-09-30, corrected book (Dominion outstanding); private-credit and venture risk from sourced assumptions.

Rebalancing options, no leverage

Without borrowing, a balanced liquid book cannot reach 15% volatility; it lands near 5%. The only unlevered way to 15% is to put the spare cash into today's mix, which keeps every imbalance and deepens the 2008 loss.

Liquid book (\$2.37M)	Volatility	Independent bets (DR ²)	BTC share of risk	Box balance (1 = equal)	2008 replay	Return if all bets are equally good / if BTC, Composer and single stocks earn no edge
Today	10.8%	3.8	41%	0.61	–20% (–\$486k)	10.2% / 5.6%
Spare cash into today's mix	15.0%	3.75	41%	0.61	–29% (–\$682k)	12.6% / 6.3%

Liquid book (\$2.37M)	Volatility	Independent bets (DR ²)	BTC share of risk	Box balance (1 = equal)	2008 replay	Return if all bets are equally good / if BTC, Composer and single stocks earn no edge
Unlevered balanced	4.9%	5.3	12%	0.83	-16% (- \$376k)	7.4% / 6.0%
Unlevered balanced, keep your metals	6.3%	4.3	7%	0.61	-18% (- \$424k)	8.0% / 6.8%

The return column is a neutral prior (cash plus $0.3 \times$ each stream's volatility), not a forecast. The two numbers carry the decision: the current book earns more only if its edge-seeking positions really have an edge. If they don't, the balanced book earns about the same at half the risk.

The unlevered balanced book holds: TIPS \$693k, long Treasuries \$284k, KMLM trend \$262k, commodities \$226k, equity ETFs \$167k, Composer \$137k as one sleeve, BTC \$65k, single stocks \$34k, ETH carry \$30k, and \$478k cash. Its four-box math puts nothing in gold, because TIPS and commodities already cover rising inflation; keeping your metals is the last row.

If you'd rather not rebuild, three measured moves each help on their own:

1. **Cap BTC at about 20% of risk** (sell about \$140k): +0.5 independent bets, and the risk spreads far more evenly.
2. **Put about \$250k of spare cash into long Treasuries**: +0.4 independent bets. The falling-growth box only reaches about 0.4% of risk; filling it takes the Treasuries and TIPS of the full rebuild.
3. **Add about \$100k to KMLM trend**: +0.15 independent bets, with correlation of 0.10 or less to everything you hold.

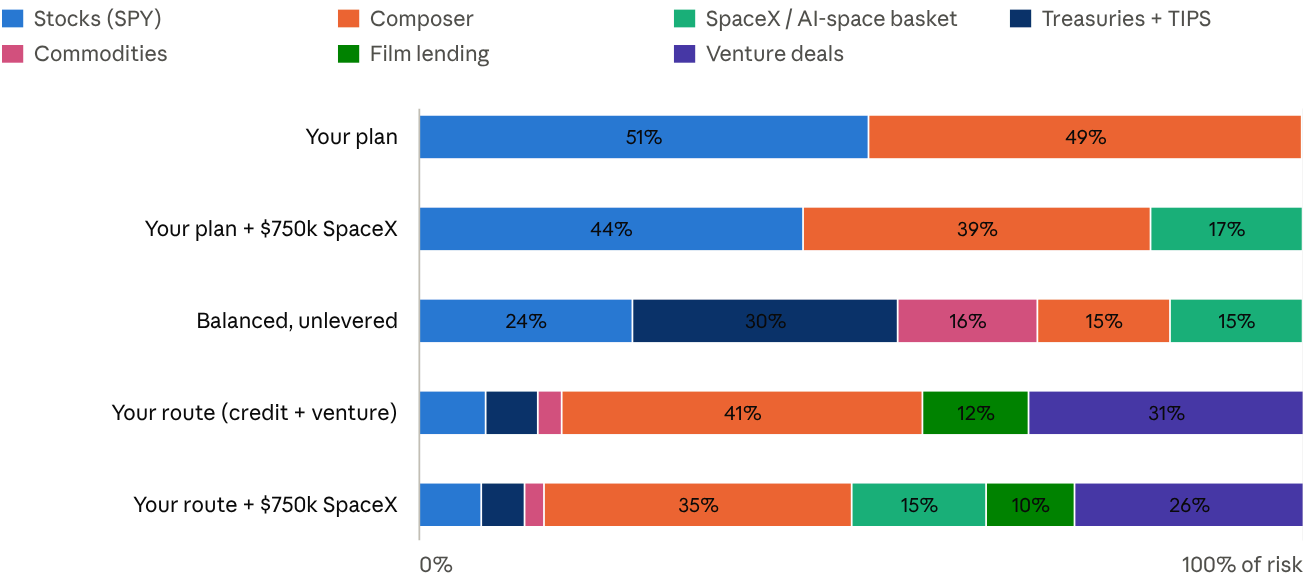
Two moves that look sensible make things worse. Shrinking the Composer HG symphony and KMLM switcher to \$120k pushes BTC to 44% of risk and balance down to 0.58. Trimming the small single stocks lowers the bet count. Taxes on selling BTC or gold are not modelled. The moves were measured on the pre-correction book; the loan correction changes cash, not risk shares.

The post-exit plan

As written, the \$18M plan is two bets, US stocks and Composer, and a \$750k SpaceX buy after a crash would add 17% of the risk on 4% of the dollars. Your own route, with private credit and venture as the return engines, holds the most independent bets and the highest expected return, but leans on Composer and on your credit and venture assumptions.

Your \$18M plan is two bets: stocks and Composer carry 100% of its risk

Share of each plan's volatility by sleeve (Euler contributions), no leverage, daily 2023-26 covariance



holygrail-engine results/casey-2026-10/no-leverage · n3_risk_vs_dollar_by_sleeve.csv · independently recomputed

Plan for the \$18M, no leverage	Volatility	Independent bets (DR²)	Box balance	Return, neutral prior	Worst replay	Bad year (1 in 20)
Your plan	10.4%	2.2	0.33	6.7%	−\$5.5M (2000–02)	−\$1.3M
Your plan + \$750k SpaceX	12.0%	2.5	0.32	7.1%	−\$6.3M (2000–02)	−\$1.9M
Balanced, unlevered	7.2%	4.0	0.73	7.4%	−\$4.3M (2008)	−\$0.4M
Your route: balanced core + film lending + venture + Composer	10.6%	4.3	0.41	9.2% (10.0% at a full 20% film rate)	−\$4.9M (2008)	−\$1.3M

Plan for the \$18M, no leverage	Volatility	Independent bets (DR ²)	Box balance	Return, neutral prior	Worst replay	Bad year (1 in 20)
Your route + \$750k SpaceX	12.0%	4.6	0.39	9.7% (10.4%)	-\$5.4M (2008)	-\$1.7M

- **Your plan earns the least.** Its \$2.88M Treasury bucket is T-bills, and the \$3.6M tech allocation waits in T-bills for the SpaceX trigger, so 36% of the dollars sit in cash with no risk or premium, while all the risk sits in stocks and Composer.
- **Your route is the closest to Dalio's idea**, because film lending and venture are different return streams. Its weak point is balance: Composer carries 41% of the risk, venture 31%, and the balanced core 16% of the risk on 47% of the dollars. The Dalio fix is to move risk budget from Composer into the core until no sleeve dominates.
- **The extra return in your route comes entirely from assumptions.** Film lending is taken at 14.4% after an allowance for defaults and extensions (20% headline), venture at the neutral prior. Your realised film income so far is 19% of plan, so this is the number to prove before scaling it to \$2.4M.
- **The balanced book has the smallest bad year** (−\$0.4M) and wins a 2000–02-style tech bust, but trails whenever US stocks lead.
- **Composer adds hidden leverage at this size.** At 24% of \$18M (\$4.32M), its 3× funds put look-through exposure at up to 1.12× (your plan) and 1.42× (your route), without any margin.

The film rate of 14.4% is $92\% \times 20\%$ minus $8\% \times$ a 50% loss. Returns are a neutral prior, not a forecast; replays hold today's weights through each window, with proxies before inception; the bad year is the 5th percentile of a 2023–26 bootstrap. Not modelled: tax on the sale, and the sale failing or pricing below \$18M.

Assumptions and open questions

Ranked by what an answer would change. Questions unanswered for 60 days from first ask are closed as refused and counted as a negative signal.

Priority	Question	What it moves	Status
P1	Which BOXX loan is still out, Dominion or Plus One, and where is the repaid loan's money now? IBKR's BOXX cost basis shows it did not go back into BOXX.	Net worth by \$150–380k; liquid book \$2.30–2.54M; liquid volatility 10–11%	Partly answered 2026-10-06 (both loans named; one still out)
P1	Your odds on Organics Ocean by July 2027: sells at \$15M or more, at \$10–15M, or slips/fails (and what it is worth if it fails). Any written banker indication?	The sale-pending section; size and timing of the \$18M plan	Asked 2026-10-06
P2	Is the \$18M before or after tax on the sale?	Size of every post-exit number	Asked 2026-10-06
P2	What triggers the SpaceX buy (for example, S&P down X% from its peak), and what size? Modelled at \$750k.	When the plan switches state; 17% of its risk	Asked 2026-10-06
P2	Are the 15–20% film rates flat per loan or annual? Typical term and extensions? How much film lending after the sale (\$2.4M modelled)?	Your route's return, 9.2–10.0%	Asked 2026-10-06
P2	Are you willing to sell physical gold and silver?	Which balanced liquid book applies	Asked 2026-10-06
P2	When will Series X (\$215k uncalled), DexMat (~\$75k) and Jarod Production (\$120k) be called, and from which account?	Liquid volatility 10.8% → 13.1% once paid	Asked 2026-10-06

Priority	Question	What it moves	Status
P3	Real estate \$1.5M: rental or residence, and any mortgage?	Whether it belongs in the book you control	Asked 2026-10-06
P2	Possible duplicates on the sheet: paper gold/silver \$68,660 vs IBKR PHYS/PSLV; RoboStrategy vs IBKR BOT; Hyperliquid inside Cash/Stables; Series X counted as venture and as uncalled cash. Each is carried once in the book, except Series X: its uncalled \$215k is counted as venture and as cash.	Net worth by up to about \$0.4M	Asked 2026-10-06
P3	Film loans: maturity dates, security, and the recovery plan for Father of Us	Default assumptions; effective loans	Asked 2026-10-06
P3	Composer VIX Harvester: its live returns track its backtest at only 0.12 correlation. Is the live symphony the one tested?	Composer risk numbers	Asked 2026-10-06
P3	BTC quantity, custody and tax basis	Cost of trimming BTC	Asked 2026-10-06

Key assumptions.

- **Liquid positions:** daily returns, trailing 3 years to 2026-09-30, with shrinkage. Composer series are backtests before each symphony went live, so their numbers are partly in-sample.
- **Private positions:** modelled from sourced assumptions, all low confidence:
 - venture at 60% volatility in the book you control (sensitivity above) and 52.5% for new post-exit deals;

- film loans with a default model;
- real estate at 15% volatility after de-smoothing appraisal marks.
- **Organics Ocean:** carried at the bankers' \$12M indication.
- **Returns:** a neutral prior (cash plus $0.3 \times \text{volatility}$), except film lending at 14.4% after defaults (20% headline) and the variants where BTC, Composer and single stocks earn no edge.
- **Not modelled:** taxes, liquidity terms, currency, and the sale as an event.

Engine, book and every result file are in `holygrail-engine/` (`books/casey-2026-10.yaml` , `results/casey-2026-10/`). Every result was independently recomputed, and this document was then cross-checked against the result files.

Holy Grail framework

Oct 6, 2026 · @Casey

Dalio's Holy Grail is one identity: a book's return-to-risk ratio equals one stream's ratio times the square root of its effective number of bets. Fifteen uncorrelated, equal-risk streams turn 6% at 18% risk into 6% at 4.65% risk, a 3.87x gain. At an average correlation of 0.2 the same fifteen are worth 3.9 bets, so the whole game is correlation and balance, not count. In real liquid markets a balanced mix holds only 2–3 independent bets, and from 2007 to 2026 balance lost to US stocks because the streams were not equally good.

What Dalio actually claims

The June 15, 2026 note makes four moves: pick good streams, make them uncorrelated, balance them by risk, then gear the mix to the risk you want. His numbers check out except one rounding slip.

Claim	His number	Exact value	Basis
One bet: 6% return, 18% standard deviation	ratio 0.3	0.333	Read today in his note
Risk with 5 / 10 / 15 uncorrelated bets	8% / 6% / 5%	8.05% / 5.69% / 4.65%	His note; recomputed
Improvement at 15 bets	4.3x (0.3 to 1.29)	3.87x, which is $\sqrt{15}$ (0.333 to 1.291)	His 4.3 is 1.29 divided by the rounded 0.3
The goal	15 good, uncorrelated, risk-balanced investments	n/a	His note
Gearing	lever the mix for much higher return at the same risk	at 18% risk: 23.2% over cash instead of 6%	His note; recomputed, borrowing at cash

Claim	His number	Exact value	Basis
Equity outlook	real returns of about -5 to -10% over 5 to 10 years	he labels it opinion; annual versus cumulative is not stated	His note
Principles (2017) and a 2011 interview	15 to 20 streams, risk down about 80%, ratio up about 5x	at 15: 74% and 3.87x; 80% and 5x need 25 streams	15-20 is the book passage, quoted second-hand; 80% and 5x are from a 2011 interview transcript; the chart itself was checked from a photo of the printed page
Chart's loss scale	ratio 0.25 means a 40% chance of a losing year; 1.25 means 11%	40.1% and 10.6% under a normal curve	Chart scales checked (10% per stream, 2.5% return); arithmetic verified

The note does not define "good", gives no correlation cutoff and no weights. He promises a strategic allocation mix "at another time".

The diversification curve

Dalio's three numbers come from one formula. Take N streams, each with volatility σ , held in equal weights, with every pair correlated ρ :

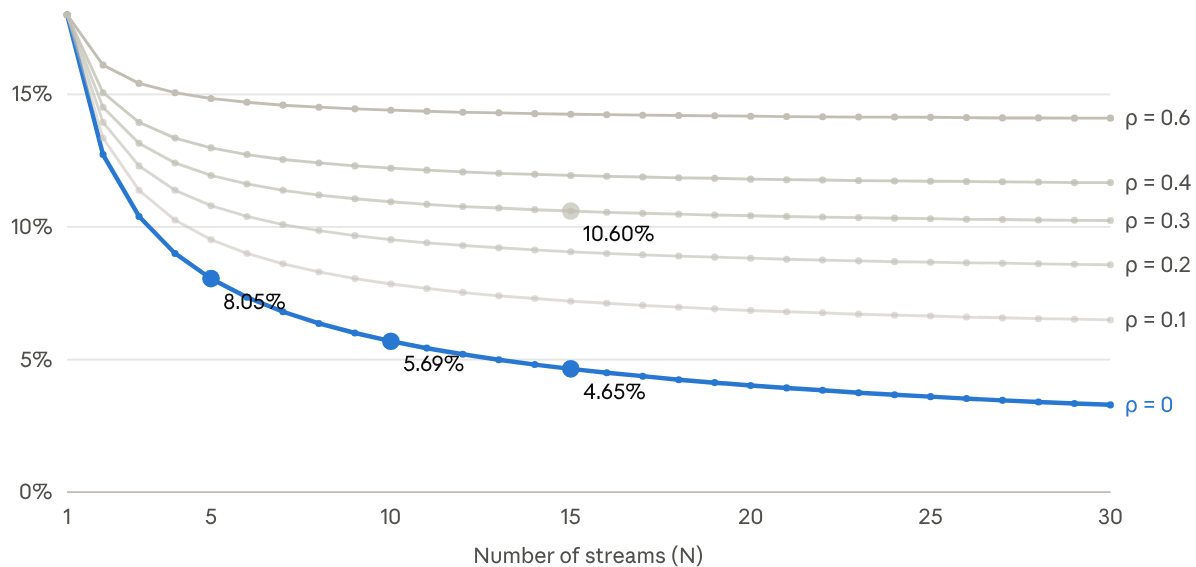
$$\sigma_p = \sigma \sqrt{\frac{1 + (N-1)\rho}{N}} \quad N_{\text{eff}} = \frac{N}{1 + (N-1)\rho} \quad \frac{S_p}{S} = \sqrt{N_{\text{eff}}}$$

15 uncorrelated bets cut risk from 18% to 4.6%; at $\rho = 0.3$, only to 10.6%

Volatility of N equal-weight streams, each 18% volatility, every pair correlated ρ

Portfolio volatility

Dalio rounds the blue points to 8%, 6% and 5%



Closed form, computed 2026-10-06 · $\sigma = 18\%$, $N = 1-30$

- **Correlation sets a floor.** As N grows, risk falls toward $\sigma\sqrt{\rho}$ and no lower: 5.7% at $\rho = 0.1$, 9.9% at $\rho = 0.3$. Past about 10 streams, another bet buys little; lower correlation buys a lot.
- **Effective bets, not count.** Fifteen streams at $\rho = 0.2$ are worth 3.95 independent bets; at $\rho = 0.3$, 2.9; at $\rho = 0.6$, 1.6.
- **Return is untouched.** Fifteen streams that each earn 6% still earn 6%; only the risk falls. The return-to-risk ratio rises by $\sqrt{N_{\text{eff}}}$, $\times 3.87$ at $N = 15$ and $\rho = 0$.

Real portfolios: risk shares and effective bets

Real streams have unequal volatilities and correlations, so the engine measures every book three ways:

$$RC_i = \frac{w_i (\sum w)_i}{\sigma_p}$$

$$DR = \frac{\sum_i w_i \sigma_i}{\sigma_p}$$

$$N_{\text{eff}} = DR^2$$

- **Risk share.** Each stream's contribution to volatility; the contributions sum to the total. A position's dollar share and its risk share usually differ a lot.
- **The diversification ratio is the Holy Grail multiplier.** If every stream has the same Sharpe S, the book's Sharpe is exactly $S \times DR$. DR^2 counts independent bets and equals $N/(1+(N-1)\rho)$ in Dalio's equal case.

- **Two kinds of "effective bets" — never mix them.** Independence measures (DR^2 , equal-p, PCA entropy) fall to 1 when streams move together. Balance measures (inverse-HHI of risk shares, minimum torsion) read 15 for fifteen equal-risk, near-identical streams. The scorecard leads with DR^2 .

11 asset classes, risk-balanced, held 1.5–5.5 independent bets — 2.3 today, not 15

Effective bets (DR^2) of the equal-risk mix, trailing 36 months of monthly returns, re-solved each month



holylrail-engine study 2026-10 · Yahoo monthly total returns 2004-12..2026-09 · VFINX VGTSX VEIEX VUSTX VFITX VIPSX VWESX VWEHX VGSIX GLD PCRIX

Over the full 2004–2026 sample the mix held 2.3 independent bets (90% bootstrap range 2.0–2.7): 2.5 in calm months, 2.2 in stress months. That is consistent with published work: a typical pension policy portfolio holds [1.2 effective bets across 7 asset classes](#), 97% of its variance from equities. Reaching 15 takes streams with different drivers — trend, carry, relative value, idiosyncratic and private bets — not more asset classes.

Gearing to a risk level

Gearing turns a better ratio into a higher return: lever the diversified mix until its volatility matches the risk you want. The book's own cash funds the extra exposure first; only real borrowing pays cash plus a spread.

$$L = \frac{\sigma^*}{\sigma_p}$$

$$g(L) = r(L) - \frac{1}{2}L^2\sigma_p^2$$

$$r(L) = r_f + L(\mu_p - r_f) - \max(L - 1, 0) s$$

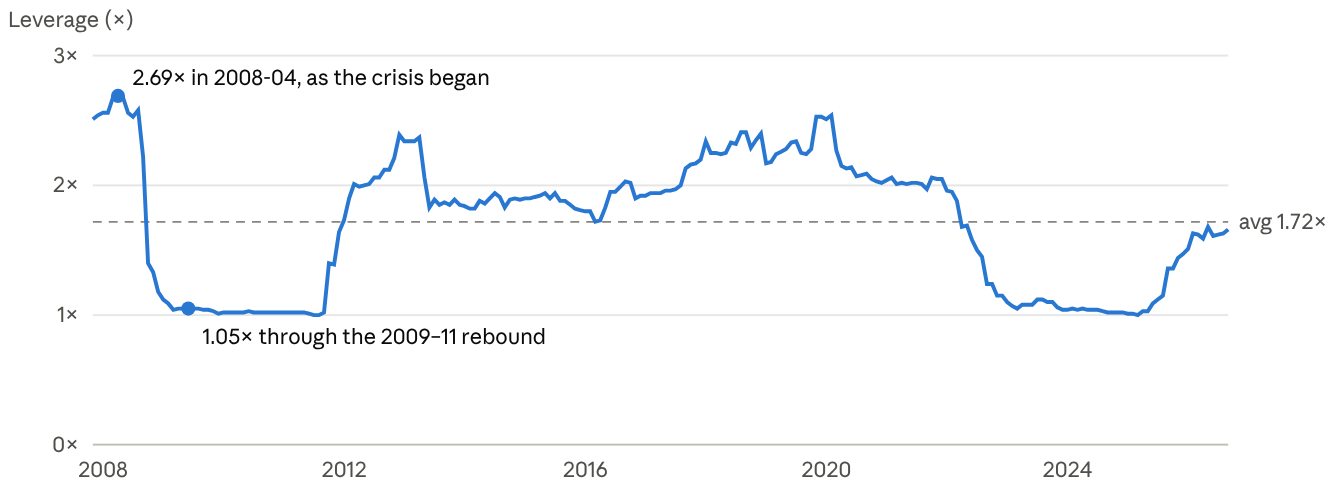
$$L_{\text{Kelly}} = \frac{\mu_p - r_f - s}{\sigma_p^2}$$

In Dalio's example, 15 streams at 4.65% volatility geared to 18% need 3.87× and earn about 23% over cash before financing, instead of 6%. Kelly says 25.5× (net of a 0.5% spread), so his gearing is only 0.15 of Kelly. The limits that bind are elsewhere:

- **Estimation error.** If the true Sharpe is half the estimate, a "Kelly" bet earns zero excess growth. A Sharpe of 0.33 needs about 35 years of monthly data to tell from zero.
- **Drawdown law.** At a fraction c of Kelly, the chance wealth ever halves is $0.5^{(2/c-1)}$: 50% at full Kelly, 12.5% at half, 0.8% at a quarter.
- **Correlation stress.** If correlation rises from 0 to 0.3, the same 3.87× book runs at 41% volatility, not 18%.
- **Financing in bad states.** Margin terms tighten exactly when assets fall.
- **How you set the leverage.** Sizing it off trailing volatility levers up after calm years, which is when risk is about to arrive.

Gearing off trailing 3-year volatility ran at 2.5–2.7x going into 2008

Leverage of the equal-risk mix geared to a 10% volatility target, walk-forward, month-end



holygrail-engine study 2026-10 · s3_leverage.csv · 36-month trailing covariance, monthly rebalance, 3× cap

The chart is the walk-forward test of that last point. Holding the same 1.72× average leverage constant would have earned 6.7% a year (Sharpe 0.48) instead of 4.3% (0.29): about 2.4 points a year lost to timing. Bridgewater sidesteps this by weighting on structural balance rather than trailing correlations, with leverage of ["around 2 times"](#).

Balance across environments

Bridgewater makes "uncorrelated" structural, not statistical. Prices move on surprises in growth and inflation, so [All Weather](#) puts equal risk on each of four surprises. It does not weight on trailing correlations, ["because they aren't stable"](#).

All Weather puts 25% of its risk on each surprise in growth and inflation

Bridgewater's four boxes; prices move on conditions relative to what is priced in

	Rising vs. expectations	Falling vs. expectations
Growth	25% of risk Equities, commodities, corporate credit, EM credit	25% of risk Nominal bonds, inflation-linked bonds
Inflation	25% of risk Inflation-linked bonds, commodities, EM credit	25% of risk Equities, nominal bonds

Most assets sit in two boxes (corporate credit in one), so the boxes are risk buckets, not asset buckets.

The All Weather Story (Bridgewater, 2012) · four boxes, 25% of risk each

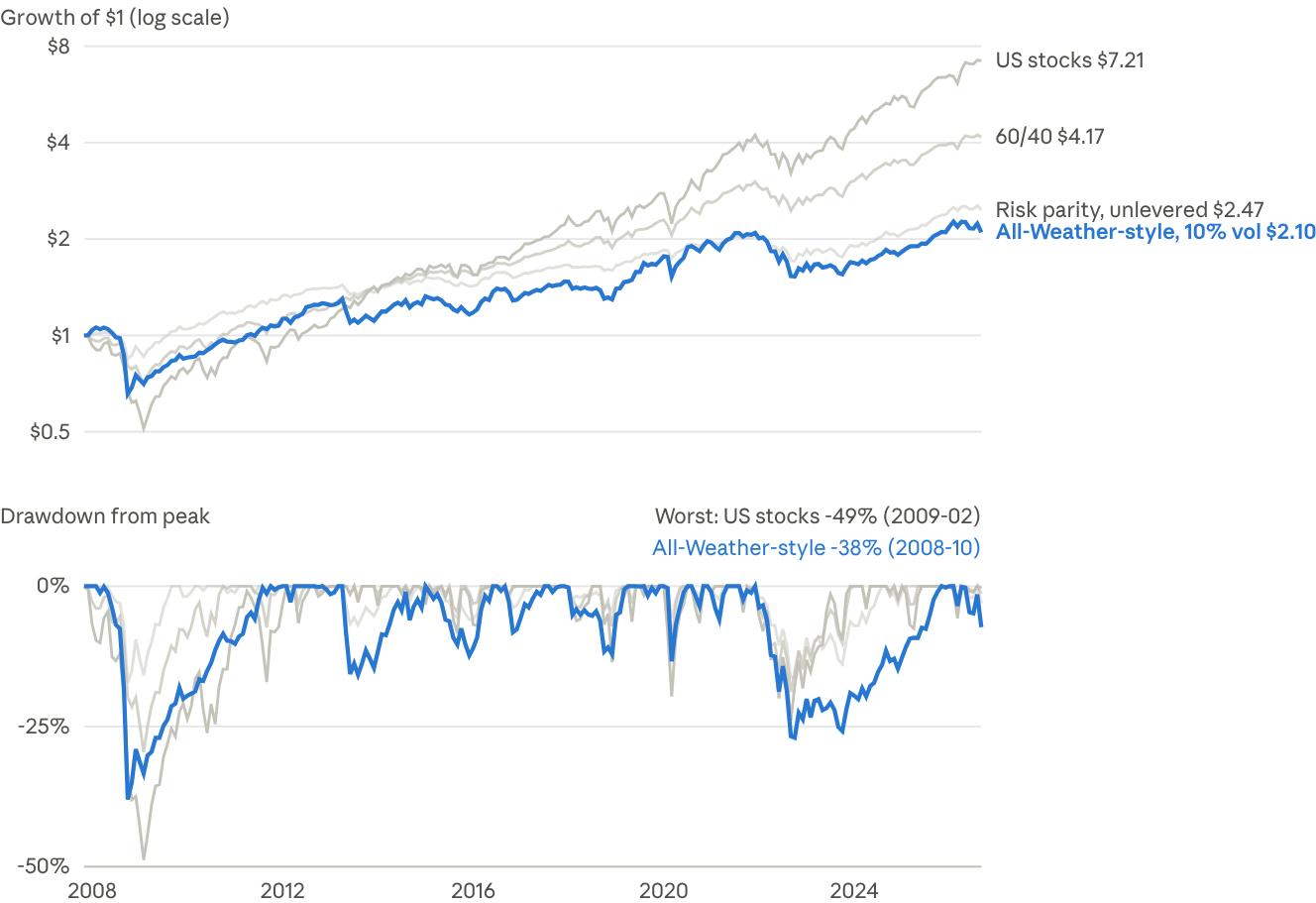
- **Lever the safe, shrink the risky.** Bonds are levered up and equities scaled down until each box carries similar risk; then the whole mix is geared to target, normally 10–12% volatility. The live [ALLW ETF](#) ran about 1.87× gross notional on 2026-10-05.
- **The stock-bond link flips with the regime.** It is negative when growth worries dominate and positive when inflation does: −0.30 in 2000–2021, +0.52 in 2022–2026. Risk-balanced stocks and bonds geared to 10% beat 60/40 in 2000–21 (Sharpe 0.82 vs 0.61), then lost 1.8% a year in 2022–26.
- **That is why the inflation boxes matter.** Without inflation-linked bonds, commodities or gold, a stock-bond book has no hedge for the 2022 kind of year. Dalio now suggests "something like ... 15%" in gold for a strategic mix (October 2025, as reported by Fortune).

What 19 years of data say

The Holy Grail's math held; its premise did not. Diversification multiplied the streams' average Sharpe by the diversification ratio, as the identity says ($0.31 \times 1.52 = 0.47$ predicted, 0.51 realised). But US stocks alone earned a Sharpe of 0.66 while the other ten streams earned 0.10–0.53, so "equally good" failed and balance lost.

2007–2026: balance lost to US stocks — \$1 became \$2.10, not \$7.21

Walk-forward, month-end, total return; the geared line pays T-bills + 0.5% on what it borrows



holygrail-engine study 2026-10 · s3_equity.csv · 11-asset universe, 36-month covariance, 10 bp costs · independently recomputed except the All-Weather-style line

Strategy, Nov 2007–Sep 2026	CAGR	Volatility	Sharpe	Max drawdown	Dec 2021–Sep 2022
US stocks	11.1%	15.6%	0.66	–48.8%	–24.0%
60/40	7.9%	9.5%	0.69	–29.6%	–18.9%
Equal weight, 11 assets	5.3%	9.8%	0.44	–30.1%	–18.8%
Risk parity, unlevered	4.9%	7.2%	0.51	–17.2%	–16.2%

Strategy, Nov 2007–Sep 2026	CAGR	Volatility	Sharpe	Max drawdown	Dec 2021–Sep 2022
Risk parity, geared to 10%	4.3%	12.5%	0.29	–38.9%	–26.8%
All-Weather-style, geared to 10%	4.0%	12.4%	0.27	–38.1%	–26.8%
Risk parity, geared to stock volatility	6.6%	17.2%	0.38	–44.2%	–43.6%

- **Unlevered balance did its job on risk.** It cut the worst drawdown to 17% from 49%, but at 7% volatility it compounded at less than half the stock rate.
- **Gearing gave the return back.** Timing the leverage off trailing volatility cost about 2.4 points a year (6.7% at constant leverage vs 4.3%), and 2022 hit the geared lines hardest (–27% against –16% unlevered), when stocks and bonds fell together.
- **It's one window.** Over 1994–2026 a risk-balanced stock/bond mix at 10% roughly matched 60/40 (Sharpe 0.67 vs 0.69), and beat it clearly in 2000–2021. Results swing with the regime and with the start date.
- **Live funds track the test.** RPAR moved with the All-Weather-style backtest (correlation 0.92, Jan 2020–Sep 2026; this tracking statistic was not independently recomputed). ALLW's 18 months are too short to judge. [Published history](#) agrees: RPAR lost 22.8% in 2022 against 15.8% for 60/40.

Basis: month-end marks, walk-forward weights, but a universe chosen in 2026; not a forecast. Month-end marks understate the geared lines' worst drawdowns. Full honesty box in `hollygrail-engine/results/study-2026-10/summary.md`.

Where it breaks

The formula is a certainty; its inputs are not. Five failure points, each one the engine tests for:

1. **Correlations rise when you need them low.** Correlation goes up in bear markets ([Longin-Solnik 2001](#)). Across 13 liquid ETFs, average pairwise correlation was 0.31 in the 2010s and 0.61 in 2022. The stock-bond sign flips with inflation uncertainty ([AQR 2022](#)). Engine: calm vs stress N_{eff} , correlation-stress scenarios.
2. **"Good" streams are scarce and hard to prove.** A Sharpe of 0.33 needs about 35 years of monthly data to tell from zero. Over 2007–2026 the stream Sharpes ranged 0.10–0.66, so the "equally good" premise failed. Engine: good-stream screen with sampling bands.

3. **Leverage has costs and timing risk.** Financed at LIBOR instead of T-bills, levered risk parity's 2.1-point edge over 60/40 (1926–2010) shrank to 0.3 points, and to –0.5 after trading costs ([Anderson-Bianchi-Goldberg](#)). Gearing off trailing volatility levered into 2008. Engine: financing spread and leverage path in every geared result.
4. **Smoothed marks fake diversification.** Appraisal and fund marks lag. With smoothing of 0.5, true volatility is 1.73× what you see and the visible beta is half the real one. Private credit, venture and real estate must be de-smoothed or modelled before they count as uncorrelated. Engine: Geltner de-smoothing, parametric private streams.
5. **Estimation error.** Sample mean-variance needs about 3,000 months of data to beat 1/N with 25 assets ([DeMiguel-Garlappi-Uppal](#)). The danger is regime change, not sampling noise: the rolling count of independent bets ranged 1.5–5.5. Engine: shrinkage, bootstrap ranges, regime splits.

Dalio's own answer to most of these is structural: balance by environment and rebalance to a fixed mix (Bridgewater, 2015). Sizing leverage so a correlation spike is survivable is our inference.

Said vs inferred

Item	What the source says	Who said it	What we added
The goal	"15 good uncorrelated investments that are risk balanced"	Dalio, 2026 note and 2025 post	The count stands in for independent bets (DR^2)
Gearing	"engineer the portfolio to the desired risk level"	Dalio, 2026	Target-vol leverage with financing on the borrowed part
"Uncorrelated"	Behaves differently by structure, not the classic correlation number	Dalio, 2011 interview (third-party transcript)	Environment mapping first; sample correlation as a diagnostic
"Good"	Roughly comparable return-to-risk (Bridgewater adds: a return above cash); "don't have to be great"	Dalio, 2011 and 2019	Screen: expected Sharpe above zero

Item	What the source says	Who said it	What we added
Four boxes	Equal risk to rising and falling growth and inflation, vs expectations	Bridgewater, 2012 and 2015	Asset-to-box mapping with sensitivities
Leverage	"around 2 times"	Bridgewater, 2011	ALLW about 1.87× gross (from its 2026-10-05 exposures)
Weights	Correlations not used in weighting "because they aren't stable"	Bridgewater, 2011	Engine offers both: structural and covariance-based
The chart	Risk scales from 10% for one stream, with return-to-risk and loss-probability scales	Principles (2017) chart, photo of the printed page	Backed out from the scales: 2.5% expected return per stream, normally distributed returns
80% less risk, 5× ratio at 15	Stated for 15 uncorrelated streams	Dalio, 2011 interview (transcript)	Exact at 15 is 74% and 3.87×; 80% and 5× need 25
Cash	"surely the worst investment over time"	Dalio, 2026	—
Gold	"something like ... 15%" of a strategic mix	Dalio, October 2025 (Fortune report)	Not in the 2012 four-box table
Equity outlook	Real returns about –5 to –10% over 5–10 years	Dalio, 2026, labelled opinion	Annual vs cumulative not stated

Not opened: the Principles book text (quotes from it are secondary), the 2011 and 2017 videos, and the primary audio behind the 2025 gold remarks.

The engine

`holygrail-engine/` in the repo runs the four operations on any book: market tickers, Composer symphonies, executor P&L, and private assets modelled from sourced assumptions. It is a library and command line, not a deployed service, so nothing needs redeploying.

Mode	What it answers	Command
Score	How many independent bets, where the risk sits, which environments are covered, how far from target volatility	<code>python -m holygrail score --book books/casey-2026-10.yaml --view liquid</code>
Backtest	Walk-forward with rebalancing, costs, financing and leverage caps; weights at t use data up to t only	<code>python -m holygrail backtest --tickers SPY,TLT,GLD,DBC,TIP --allocator erc --target-vol 0.10</code>
Forward	Monte Carlo (normal, fat-tailed, block bootstrap), correlation stress, 2008 / 2020 / 2022 replays	<code>python -m holygrail forward ... / stress ...</code>
Environments	Growth × inflation regimes from FRED, risk per box	<code>python -m holygrail envs ...</code>
Curve	Dalio's closed form	<code>python -m holygrail curve</code>

- **Allocators:** equal weight, inverse volatility, equal risk and custom risk budgets, minimum variance, maximum diversification, HRP, four-box balance, target-volatility gearing.
- **Books are honest by construction.** Every assumption carries a value, source and confidence; a bare number is refused, and totals must reconcile to the dollar.
- **Verified.** 386 gate tests pass offline. Two independent reviews found 33 issues (2 blocking, 9 serious); all real ones are fixed with regression tests. The last fix, cash earning its own rate inside a geared book, was checked against brute force across 684 cases.

Sources

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